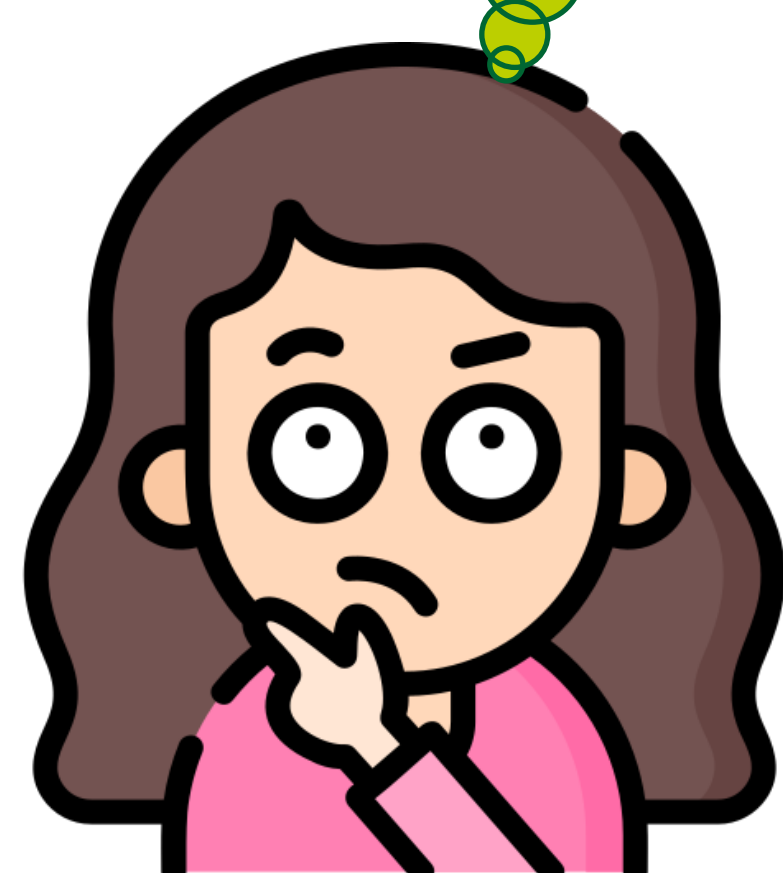


Hexavalent vaccine: does the 2+1 schedule increase *Haemophilus influenzae* type b cases?

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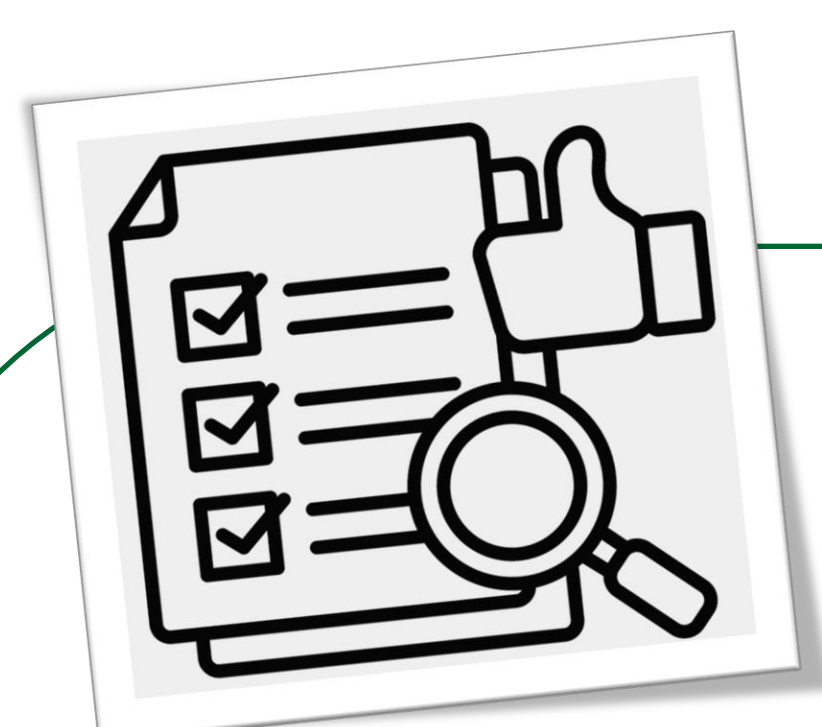
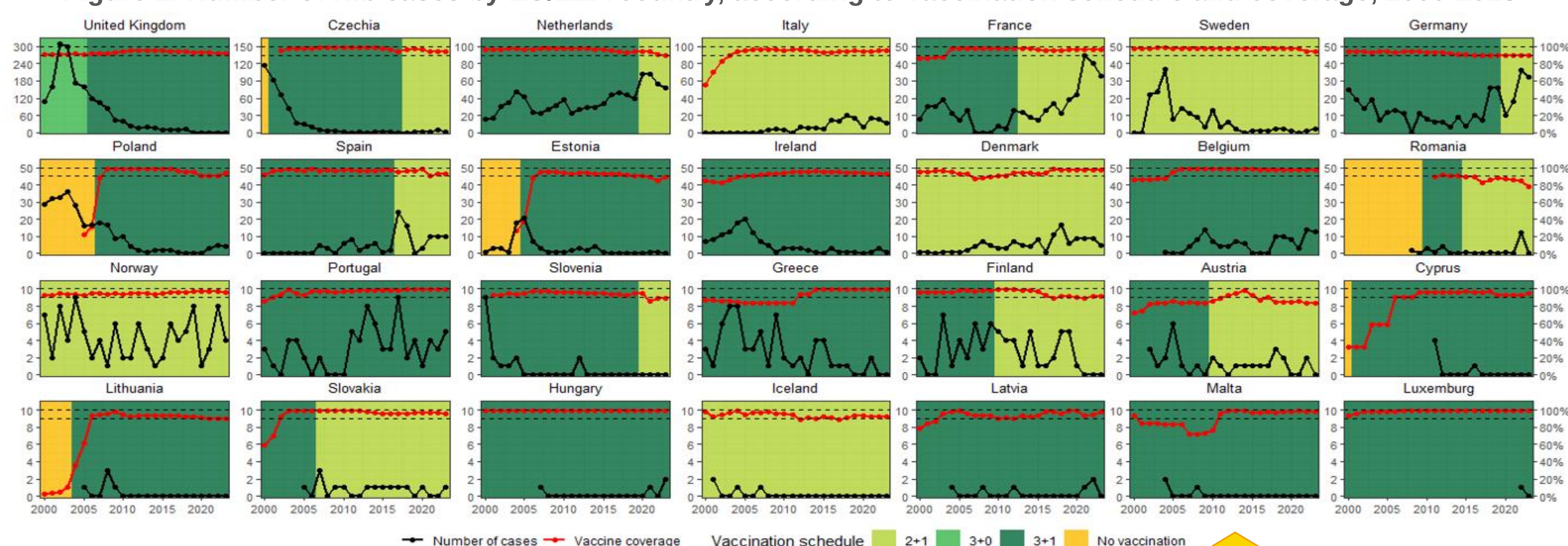
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3+1 or 2+1
schedule?



- Belgian National Immunization Technical Advisory Group (NITAG) is considering shifting from 3+1 to 2+1 hexavalent vaccine schedule.
- In some countries as France and the Netherlands, with a switch to 2+1 schedule, an increase of *Haemophilus influenzae* type b (Hib) is observed.
- This study compares Hib trends and vaccination schedules across the EU/EEA to explore potential causes of such increase.
- Data retrieved from official sources (ECDC, WHO).

Figure 1: Number of Hib cases by EU/EEA country, according to vaccination schedule and coverage, 2000-2023



Key Findings

- **16/28 countries** use a 2+1 schedule.
- Some countries experienced an increase but it appears **before** (Germany), **immediately** after (the Netherlands) or **more than a decade** (Italy) after introduction of 2+1 schedule (Figure 1).
- Some countries with 2+1 schedule **do not report increase** (Slovakia, Austria, Nordic countries).
- For some countries with increase, vaccine coverage is **less than 90%** (e.g. Germany and the Netherlands).
- In **France**, the increase in cases occurred a few years after the introduction of the 2+1 schedule and is not linked to a reduction in vaccination coverage.
- In countries with 2+1 schedule, **age of booster** is **variable**. Countries with increase of cases administer booster at 11 months.
- **No country has reverted** to a 3+1 schedule.

- **Increase of Hib is probably multifactorial.**
- **In addition to vaccination schedule, coverage, type of vaccine and age for booster, further elements could be explored as quality of surveillance and attendance of childcare at young age.**

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